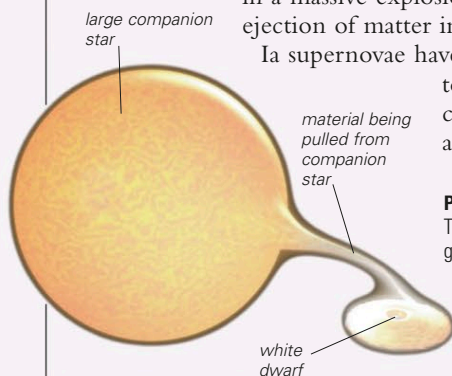


## TYPE I SUPERNOVAE

As in a nova (opposite), the source of a type I supernova is a binary system consisting of a giant star and a white dwarf. In type I supernovae, rather than triggering a nova, the material transfer onto the white dwarf continues to increase the mass of the star until it collapses and then explodes, destroying itself. The class of type I supernovae is subdivided depending on which chemical elements are present in the supernova's spectrum. In type Ia supernovae, the core of the white dwarf reaches a critical density, triggering the fusion of carbon and oxygen. This fusion is unconstrained and results in a massive explosion, with an associated leap in luminosity and the ejection of matter into interstellar space. According to theory, all type Ia supernovae have identical luminosities. This means that the distance to a supernova of this type can be determined by comparing its intrinsic luminosity with its apparent brightness.



### POWERFUL SUPERNOVA

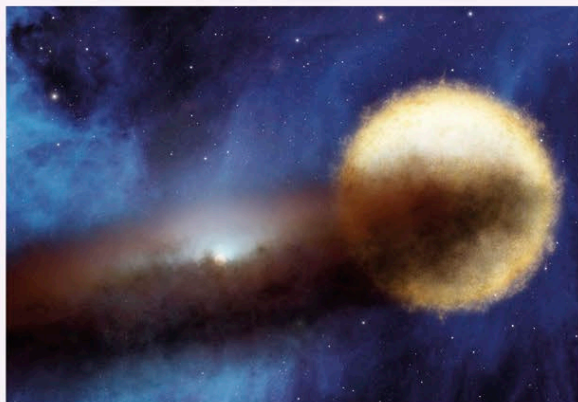
This white-dwarf star pulls gas from a larger companion. Its mass rises until it can no longer support itself and it collapses in a huge explosion.

Supernova 1994D



### DISTANT SUPERNOVA

Like other type Ia supernovae, 1994D—seen in the outskirts of the distant galaxy NGC 4526—has an intrinsic brightness that allows its distance to be known.



### MYSTERIOUS STAR

One of the strangest stars known to astronomers, Epsilon Aurigae is a giant star that is being eclipsed by something even bigger than itself. One theory is that it is being orbited by a large, dusty disk surrounding a binary companion.

## BIZARRE VARIABLES

Many variable stars exhibit magnitude variations that are regular and are easily explained by eclipsing or by a pulsation mechanism occurring in a star's outer layers. However, there are other variable stars whose magnitude variations seem to defy explanation. One example is Epsilon (ε) Aurigae or Almaaz, a giant star with eclipses that last for 2 years, far longer than expected for a normal eclipsing system. As Almaaz is itself huge, whatever is eclipsing it must be even larger, but in the absence of decisive observations, astronomers can only theorize. One theory is that there is an unseen companion star or stars surrounded by a large dust ring, and it is the extended dust ring that eclipses Almaaz. Another bizarre variable is R Coronae Borealis (R CrB). This star can suddenly drop eight magnitudes, a large range that cannot be explained by physical changes within the star's structure (see p.287). The variation cannot be due to an eclipse, since the drop in magnitude is irregular and not periodic. Some astronomers have suggested that an orbiting dust cloud is responsible, but the more popular theory is that R CrB is ejecting material from its surface and that this ejected material blocks the light from the star before being blown away. Although the majority of variable stars are well understood, even to the extent that they can be used as reliable distance indicators, there are many individual stars that require further study before they reveal their secrets.



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